



Odysseus

Kali Linux Setup Guide

A practical step-by-step guide for installing and running the Odysseus project using Docker Compose.

Repository

<https://github.com/odysseus-dev/odysseus>

This guide uses the **main** branch and keeps the installation as close to the repository's default configuration as possible.

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1 Requirements

Recommended environment:

- Kali Linux
- Git
- Docker
- Docker Compose
- Internet connection
- At least several GB of free disk space
- 8 GB RAM or more recommended

Odysseus runs through Docker, so Python dependencies do not need to be installed directly on the Kali host.

2 Check Docker

First verify Docker:

```
docker --version
```

Then verify Docker Compose:

```
docker compose version
```

Example:

```
Docker version 29.4.2  
Docker Compose version v5.1.3
```

If both commands work, continue.

3 Clone the Repository

Create/use a Projects directory:

```
cd ~/Desktop/Projects
```

Clone the **main** branch:

```
git clone --branch main --single-branch https://github.com/odysseus-dev/odysseus.git
```

Enter the repository:

```
cd ~/Desktop/Projects/odysseus
```

Confirm the branch:

```
git branch --show-current
```

Expected:

```
main
```

Optionally check the working tree:

```
git status --short
```

A clean checkout should produce no output.

4 Create the Environment File

Odysseus provides an example environment file.

Create the actual `.env` file:

```
cp .env.example .env
```

Do not delete `.env.example`. The `.env` file contains environment-specific configuration and should generally not be committed to Git.

5 Important Default Ports

Odysseus uses several services.

Service	Container Port	Host Port
Odysseus	7000	7000
ChromaDB	8000	8100
SearXNG	8080	8081
ntfy	80	8091

Important distinction: SearXNG uses port 8080 inside Docker, but its host-side port is changed to 8081 in this setup.

6 Check for Port 8080 Conflicts

Before starting the project, check whether port 8080 is already occupied:

```
sudo ss -ltnp 'sport = :8080'
```

If another service is already using 8080, do **not** stop or modify that service blindly.

In this setup, port 8080 was already being used by an existing database service.

The existing database on port 8080 was important.

7 Resolve the SearXNG Port Conflict

The repository's original Compose configuration maps SearXNG like this:

```
ports:
  - "127.0.0.1:8080:8080"
```

That means Docker attempts to expose SearXNG on host port 8080. If host port 8080 is already occupied, Docker Compose will fail with an error similar to:

```
Bind for 0.0.0.0:8080 failed: port is already allocated
```

Change **only** the host-side SearXNG port:

```
sed -i 's/127\.\.0\.\.0\.\.1:8080:8080/127.0.0.1:8081:8080/' docker-compose.yml
```

The resulting mapping is:

```
ports:
  - "127.0.0.1:8081:8080"
```

This means:

```
Kali host:8081 -> SearXNG container:8080
```

Do not change SEARXNG_INSTANCE=http://searxng:8080. Odysseus communicates with SearXNG through Docker's internal network, where SearXNG remains available at http://searxng:8080.

8 Start Odysseus

Start the complete stack:

```
docker compose up -d
```

A successful startup should show containers for:

```
odysseus
chromadb
searxng
ntfy
```

SearXNG should become healthy before Odysseus fully starts.

9 Check the Containers

Check the running containers:

```
docker compose ps
```

The important containers are:

```
odysseus-odysseus-1
odysseus-chromadb-1
odysseus-searxng-1
odysseus-ntfy-1
```

All should be running.

10 Access the Odysseus Web Interface

Open:

```
http://localhost:7000
```

The Odysseus login page should appear.

11 Configure the Admin Account

Odysseus uses the following settings in `.env`:

```
ODYSSEUS_ADMIN_USER  
ODYSSEUS_ADMIN_PASSWORD
```

For example:

```
ODYSSEUS_ADMIN_USER=your_username  
ODYSSEUS_ADMIN_PASSWORD=your_password
```

11.1 Password requirement

The setup script requires the password to be at least **8 characters**. A password shorter than 8 characters causes:

```
[error] ODYSSEUS_ADMIN_PASSWORD must be at least 8 characters
```

Use a strong password of at least 8 characters.

12 Important: auth.json Persistence

Odysseus stores the initialized authentication configuration in:

```
/app/data/auth.json
```

The setup flow is approximately:

```
auth.json does not exist  
  |  
  v  
create admin from ODYSSEUS_ADMIN_USER/PASSWORD  
  |  
  v  
auth.json created
```

On subsequent starts:

```
auth.json exists  
  |  
  v  
skip admin creation  
  |  
  v  
existing credentials remain
```

Therefore, changing `.env` after the admin account has already been created does **not necessarily change the existing login credentials**.

13 If Login Says “Invalid Credentials”

Do not immediately delete files.

First verify the environment variables inside the running container:

```
docker compose exec odysseus printenv | grep -E '^ODYSSEUS_ADMIN_(USER|PASSWORD)='
```

The values should correspond to the credentials configured in `.env`.

If the credentials were configured incorrectly during first initialization, inspect the setup process:

```
docker compose exec odysseus sh -lc 'python /app/setup.py'
```

The setup script will tell you whether:

- the admin account already exists;
- the password is too short; or
- setup completed successfully.

14 Do NOT Delete `auth.json` Unless You Intend to Reinitialize Authentication

The authentication file is:

```
/app/data/auth.json
```

Deleting it causes Odysseus to recreate the initial admin account during setup. This should only be done deliberately when you actually need to reinitialize authentication.

If an existing installation has working credentials, **leave `auth.json` alone**.

15 Understanding the Startup Process

Odysseus uses the following startup chain:

```
Docker
|
v
Docker Compose
|
v
Odysseus container
|
v
entrypoint.sh
|
v
setup.py
|
v
Uvicorn
|
v
Odysseus application
```

The Docker image uses:

```
ENTRYPOINT ["/usr/local/bin/entrypoint.sh"]
```

and ultimately starts Uvicorn on:

```
0.0.0.0:7000
```

The entrypoint also runs `/app/setup.py` during container startup. The setup script is designed to be idempotent, meaning it can run repeatedly without recreating an existing admin account or unnecessarily rebuilding initialized data.

16 Automatic Startup After Reboot

The containers in this setup use the Docker restart policy:

```
unless-stopped
```

The four containers are configured as:

```
odysseus-odysseus-1      unless-stopped
odysseus-chromadb-1     unless-stopped
odysseus-searxng-1      unless-stopped
odysseus-ntfy-1         unless-stopped
```

Therefore, once Docker itself starts after a Kali reboot, these containers automatically restart.

You do **not** need to manually run `docker compose up -d` after every reboot. The condition is that the Docker service itself must start automatically.

17 Verify Restart Policy

If you ever want to verify the restart policy:

```
docker inspect -f '{{.Name}} {{.HostConfig.RestartPolicy.Name}}' odysseus-odysseus-1
odysseus-chromadb-1 odysseus-searxng-1 odysseus-ntfy-1
```

Expected:

```
/odysseus-odysseus-1 unless-stopped
/odysseus-chromadb-1 unless-stopped
/odysseus-searxng-1 unless-stopped
/odysseus-ntfy-1 unless-stopped
```

18 Check Odysseus Logs

To view Odysseus logs:

```
docker compose logs odysseus
```

For live logs:

```
docker compose logs -f odysseus
```

Successful startup should eventually contain messages similar to:

```
Application startup complete
Uvicorn running on http://0.0.0.0:7000
```

19 What Successful Startup Looks Like

During the first startup, Odysseus may initialize:

- FastEmbed
- ChromaDB collections
- Vector RAG
- MemoryVectorStore
- TTS
- STT
- MCP routes
- Background-job monitoring
- Task scheduler
- Built-in MCP servers

Messages about collections initially returning 404 during first-time initialization are not necessarily errors. A collection may be queried before it exists and then created. The important condition is that startup eventually reaches **Application startup complete**.

20 ChromaDB

ChromaDB runs inside Docker.

Internal container port	8000
Host-side port	8100

Odysseus connects internally using:

```
CHROMADB_HOST=chromadb
CHROMADB_PORT=8000
```

Do not change these to `localhost` when Odysseus is running inside Docker. Inside Docker, `odysseus` -> `chromadb:8000` is correct.

21 SearXNG

SearXNG runs internally on:

```
searxng:8080
```

Its host-side port in this setup is:

```
localhost:8081
```

Therefore:

```
Browser -> localhost:8081
        -> SearXNG container -> port 8080
```

Odysseus itself continues to use:

```
http://searxng:8080
```

Do not confuse the host port with the Docker-network port.

22 ntfy

ntfy is exposed on:

```
localhost:8091
```

Its container listens on port 80. The mapping is:

```
8091 -> 80
```

23 Odysseus Main Interface

The main application is:

```
http://localhost:7000
```

The default Compose binding is loopback:

```
127.0.0.1:7000
```

This means it is intended to be accessed locally from the Kali machine unless you deliberately change the binding.

24 Authentication

The relevant defaults include:

```
AUTH_ENABLED=true  
LOCALHOST_BYPASS=false
```

This means authentication is enabled. Do not disable authentication unless you deliberately understand the security consequences. For a normal local installation, keeping authentication enabled is the safer configuration.

25 Database

By default the application can use:

```
DATABASE_URL=sqlite:///./data/app.db
```

The Docker Compose configuration also supports overriding the database URL through `.env`.

Do not change the database configuration unless you actually need another database. In particular, if the Kali machine already has another database service using host port 8080, do not stop or reconfigure it simply to make SearXNG use 8080.

26 GPU Considerations

A machine may have an NVIDIA GPU and working `nvidia-smi` while Docker still lacks an NVIDIA container runtime. In that case, the normal CPU-based Compose configuration is appropriate.

Do not install NVIDIA Container Toolkit merely because the machine has an NVIDIA GPU. Only add GPU Docker support if GPU acceleration is actually required and you have verified that the hardware, drivers, CUDA stack, and Odysseus GPU Compose configuration are appropriate.

27 Disk Space

Odysseus builds a fairly large Docker image because of its Python/AI dependencies. The build can take considerable time on a lower-powered laptop.

During the installation documented here, the Docker build took roughly 17 minutes. Do not assume that a long build means the process is stuck. Check the Docker output before interrupting it.

28 Optional TTS Cache Warning

The Compose configuration may produce a warning about `ODYSSEUS_TTS_CACHE_MAX_BYTES` being unset. This is an optional configuration.

If the application starts successfully, the warning does not prevent Odysseus from running. There is no need to invent a value unless you specifically want to configure the TTS cache size.

29 Useful Daily Commands

29.1 Start the stack manually

```
cd ~/Desktop/Projects/odysseus
docker compose up -d
```

29.2 Stop the stack

```
docker compose down
```

29.3 Restart Odysseus

```
docker compose restart odysseus
```

29.4 View all containers

```
docker compose ps
```

29.5 View Odysseus logs

```
docker compose logs odysseus
```

29.6 Follow Odysseus logs

```
docker compose logs -f odysseus
```

30 Updating the Repository

Before updating, check your branch:

```
git branch --show-current
```

It should say:

```
main
```

Check for local changes:

```
git status --short
```

Then update:

```
git pull origin main
```

After pulling a new version, rebuild/recreate containers if the project changes require it:

```
docker compose up -d --build
```

Do not blindly delete Docker volumes or application data when updating.

31 Critical Things NOT to Do

31.1 Do not occupy existing port

If another service already uses a port, leave it alone. Use another host port for SearXNG. This setup uses 8081 for SearXNG. You do not delete/override any ports unless you are sure.

31.2 Do not delete application data unnecessarily

Do not randomly remove `/app/data` or Docker volumes. They may contain authentication data, database data, vector data, application state, and other persistent information.

31.3 Do not delete `auth.json` casually

`/app/data/auth.json` contains the initialized authentication configuration. Only remove it intentionally when resetting the initial authentication setup.

31.4 Do not change internal Docker hostnames to `localhost`

Inside the Docker network, `searxng:8080` and `chromadb:8000` are correct. Changing them to `localhost` would point to the Odysseus container itself rather than the other containers.

32 Troubleshooting

32.1 Error: port 8080 already allocated

You may see:

```
Bind for 0.0.0.0:8080 failed: port is already allocated
```

Check the port:

```
sudo ss -ltnp 'sport = :8080'
```

If an important existing service uses it, do not stop it. Change only the SearXNG host mapping to:

```
127.0.0.1:8081:8080
```

32.2 Error: Invalid credentials

Check the configured admin environment:

```
docker compose exec odysseus printenv | grep -E '^ODYSSEUS_ADMIN_(USER|PASSWORD)='
```

Then check setup:

```
docker compose exec odysseus sh -lc 'python /app/setup.py'
```

If `auth.json` already exists, the existing authentication database is being preserved. If you intentionally need to recreate the initial admin account, remove `auth.json` and restart/reinitialize the application.

32.3 Error: Password must be at least 8 characters

The setup script requires a minimum password length of 8. Use a password with at least 8 characters.

32.4 Odysseus container keeps restarting

Check:

```
docker compose logs odysseus
docker compose ps
```

Look for the first actual error rather than only the final restart message.

32.5 SearXNG is unhealthy

Check:

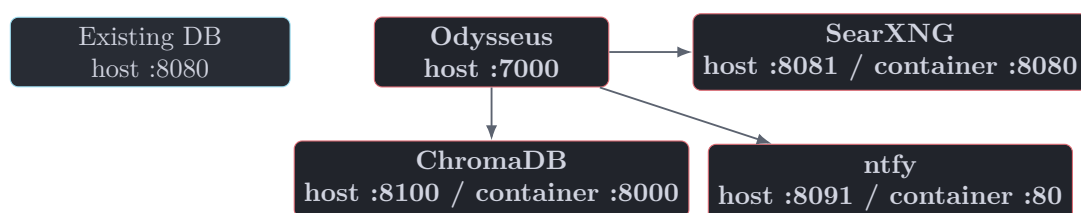
```
docker compose logs searxng
```

Remember:

```
host = 8081
container = 8080
```

The host port does not need to match the container port.

33 Final Port Layout



Inside the Docker network:

```
Odysseus -> searxng:8080
Odysseus -> chromadb:8000
```

The existing host database remains on :8080 and is not modified.

34 Final Verification

A correctly installed system should have:

```
Git branch:
main

Odysseus:
localhost:7000

SearXNG:
localhost:8081

ChromaDB:
localhost:8100

ntfy:
localhost:8091

Existing DB:
localhost:8080

Restart policy:
unless-stopped
```

The four Odysseus containers should report:

```
odysseus-odysseus-1      unless-stopped
odysseus-chromadb-1     unless-stopped
odysseus-searxng-1      unless-stopped
odysseus-ntfy-1         unless-stopped
```

Once Docker starts after a reboot, the Odysseus containers automatically restart.

35 Minimal Installation Checklist

For a new Kali user, the shortest installation path is:

```
cd ~/Desktop/Projects
git clone --branch main --single-branch https://github.com/odysseus-dev/odysseus.git
cd ~/Desktop/Projects/odysseus
cp .env.example .env
sudo ss -ltnp 'sport = :8080'
sed -i 's/127\.\.0\.\.0\.\.1:8080:8080/127.0.0.1:8081:8080/' docker-compose.yml
docker compose up -d
```

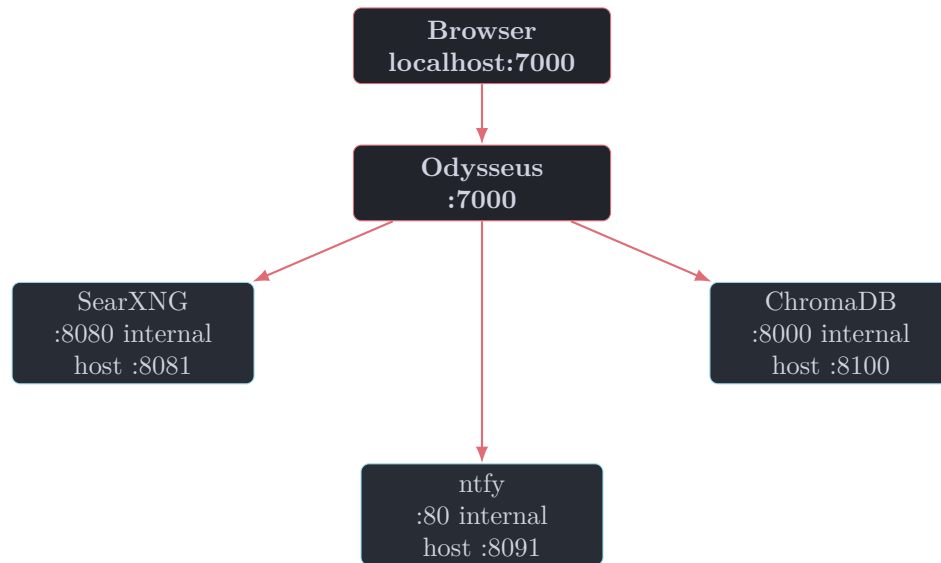
The SearXNG mapping command should only be run when the original host 8080 mapping conflicts with an existing service. If the repository configuration has already been changed, do not blindly run the substitution again.

Then open:

```
http://localhost:7000
```

Create/use an admin password of at least 8 characters.

36 Final Architecture



The key principle is:

Only the host-side SearXNG port was changed. The existing database on port 8080 was left untouched.

With Docker's `unless-stopped` restart policy, the installation is persistent across normal system reboots, assuming Docker itself starts automatically.

End of guide